Karthik Soma

Code ◆ linkedin ◆ E-mail ◆ Gscholar

Education

Polytechnique Montréal, Mila Quebec AI Institute — *Sept 2021 – Dec 2026 (expected)*
Ph.D., Department of Computer and Software engineering — *GPA: 3.98 / 4.00*
Advisor: Prof Giovanni Beltrame

National Institute of Technology, Tiruchirappalli — *Jul 2017 – Jul 2021*
B.Tech in Instrument and Control engineering — *GPA: 8.98 / 10.00*

Publications

Accepted

Multi-Robot Decentralized Collaborative SLAM in Planetary Analogue Environments: Dataset, Challenges, and Lessons Learned — Arxiv, Code
PY Lajoie, K Soma, HM Bong, A Lemieux-Bourque, R Zhang, VS Varadharajan, G Beltrame
IEEE Transactions on Field Robotics (T-FR) / ICRA, 2026

Split Over n Resource Sharing Problem: Are Fewer Capable Agents Better Than Many Simpler Ones? — Arxiv
K Soma, MS Talamali, G Miyauchi, G Beltrame, H Hamann, R Groß
International Conference on Swarm Intelligence (ANTS), 2026

Sociodynamics of Reinforcement Learning — Arxiv, Code
Y Bouteiller, K Soma, G Beltrame
Transactions on Machine Learning Research (TMLR) / RLC, 2026

The Heterogeneous Multi-Agent Challenge — Arxiv, Code
C Dansereau, JS Lopez-Yepez, K Soma, A Fagette
European Conference on Artificial Intelligence (ECAI), 2025

A Complete Set of Connectivity-aware Local Topology Manipulation Operations for Robot Swarms — Arxiv
K Soma*, K Khateri*, M Pourgholi, M Montazeri, L Sabattini, G Beltrame
IEEE International Conference on Robotics and Automation (ICRA), 2023

Congestion and Scalability in Robot Swarms: A Study on Collective Decision Making — Arxiv, Code
K Soma, VS Vardharajan, H Hamann, G Beltrame
International Symposium on Multi-Robot and Multi-Agent Systems (MRS), 2023

Collective Transport via Sequential Caging — Arxiv, Code
VS Vardharajan, K Soma, G Beltrame
International Symposium on Distributed Autonomous Robotic Systems (DARS), 2021

Preprints

The Hive Mind is a Single Reinforcement Learning Agent — Arxiv, Code
K Soma, Y Bouteiller, H Hamann, G Beltrame
ArXiv preprint, 2025

Guided by Guardrails: Control Barrier Functions as Safety Instructors for Robotic Learning — Arxiv
M Guerrier, K Soma, H Fouad, G Beltrame
ArXiv preprint, 2025

Can Vision Foundation Models Navigate? Zero-Shot Real-World Evaluation and Lessons Learned — Arxiv, Code
M Guerrier, K Soma, J Pavlasek, G Beltrame
ArXiv preprint, 2025

Hierarchies Define the Scalability of Robot Swarms — Arxiv
VS Varadharajan, K Soma, S Dyanatkar, PY Lajoie, G Beltrame
ArXiv preprint, 2024

Recent Projects

Mitacs Accelerate - Thales Digital Solutions Inc. | Research Internship | Montréal, Canada *Aug 2024 – Mar 2026*
Python, PyTorch, ROS, Gazebo, Multi-Agent Reinforcement Learning

- Benchmarked various MARL algorithms in complex coordination tasks involving heterogeneous agents.
- Developed *CamOnlyReachHome*: a camera-only goal-conditioned homing system for drones to handle scenarios when the robot is in localization failure mode.

Robot Systems for Space Technologies | Field Robotics, Multi-Robot Systems *Sept 2021 – present*
ROS, Rovers, Collaborative SLAM, Boston Dynamics Spot

- Developed a multi-robot system for the ESA/ESRIC Prospecting Challenge to explore and prospect rock samples in a lunar analogue environment in the Netherlands; responsible for local planning and delay management modules.
- Led navigation for a Boston Dynamics Spot robot in the *Save the Astronaut* project, a cave exploration mission.
- Contributed to a Canadian Space Agency (CSA) Mars-analogue collaborative SLAM dataset, later covered by CNN.

Swarm Biotactics | Bio-hybrid Robotics, Swarm Intelligence *Dec 2025 – present*
Collaborative SLAM, Collective Navigation

- Developed a Argos3 based simulator for testing localization and mapping algorithms on low compute devices.
- Designing collective strategies for centroid estimation, flocking and collective navigation.

Skills

Programming Languages: LaTeX, C/C++, Embedded C, MATLAB, Buzz and Python
Frameworks: PyTorch, Robotics Operating System (ROS) and Docker
Hardware: Raspberry Pi, Arduino (Uno, Mega, Nano), ESP8266, NVIDIA Jetson (Nano, Xavier, Orin)
Simulation Software: Gazebo, Argos3, Unity and CoppeliaSim
Robots: Ground (Agilex Limo, Rover & Bunker; Khepera; Duckiebot), Aerial (Crazyflie; Cognifly), Legged (Boston Dynamics Spot; Unitree A1 & GO2), Humanoid (Unitree G1)

Talks

The Hive Mind is a Single Reinforcement Learning Agent *2026*
D&M Mila Annual Conference, Montréal, Canada (Jan 2026); University of Konstanz, Germany (Jun 2026)

Achievements

- Awarded Mitacs Accelerate grant of 45,000 CAD for research internship at Thales Digital Solutions Inc.
- Awarded Mitacs grant for 6000 CAD to conduct research on Split Over *n* Resource Sharing Problem at Sheffield University.
- Awarded travel grant of 3,000 CAD for ICRA'23 at London.
- Selected to participate in the IEEE MRS Summer School'22 in Prague, Czech Republic.
- Awarded Amii talent bursary of 1500 CAD to attend Amii Talent Week'22 in Edmonton, Alberta, Canada.
- Secured 1st Prize in Daksh'20 for Smart Manufacturing Hackathon at Shaastra University, Thanjavur.
- Secured 4th out of 100 teams in the National level Robotics Competition conducted by IIT Bombay (E-Yantra).

Positions of Responsibilities

- Organizing the Multi-Robot Summer School (MRSS) 2026 in Montréal, Canada.
- Teaching Assistant for INF6805E: Swarm Intelligence'26, Polytechnique Montréal.
- Held various roles at National Institute of Technology, Tiruchirappalli (NIT-T)

- Inter Profile Co-ordinator of Spider (R&D) club.
- Workshop Organiser, Sensors'19 & Sensors'20, National department symposium of ICE dept.
- Member of teaching clubs Ignitte & U and I.
- Deputy Manager, Festember Media Relations team, Festember'19.
- Winning Hostel Coordinator, freshman intra-hostel competition Aaveg'18.